Reinforcement Learning Is All You Need

Next-Token Prediction as a Special Case of Reward Maximization

Lemmata

Abstract

Today's language models are trained in two stages that are treated as two different kinds of learning: next-token prediction (NTP) on text, then reinforcement learning (RL) on rewards. We show that the first stage is already the second. When the model's action is its distribution over the next token and the reward is a strictly proper scoring rule, reward regret equals prediction error exactly, and with the logarithmic score the RL gradient *is* the familiar cross-entropy gradient: NTP is reinforcement learning with the variance taken out. Through Solomonoff's theorem, RL thereby inherits the strongest known guarantee of universal prediction, and we prove that this survives in the format language models actually use, sampled tokens, as long as each reward looks at two samples rather than one. The core of the argument is machine-checked in Lean 4. We then put the idea to work. Transformers trained from nothing but scalar rewards learn to predict text and Lean code at the same compute per step as NTP. And with the Lean proof checker as the only teacher, a 30M-parameter model that never saw a proof learns to solve 86 of 212 miniF2F competition problems using 3.7 verifier calls per problem, reaching 97% of a strong automation baseline at about a third of its cost. The result is a single principle for building language models: one reward-maximization objective, in which pretraining is simply the densest reward, and verifiers, tests and environments supply the rest.

Keywords: reinforcement learning, next-token prediction, proper scoring rules, Solomonoff induction, language models, formal theorem proving, Lean

At a glance

- **One principle.** Next-token prediction is the special case of reinforcement learning in which the reward is a proper score on data and the expectation over actions is taken exactly (Section 3).
- **Universal by construction.** A single RL policy has finite cumulative regret on every prediction task generated by any computable source; with two-sample rewards this holds for token-sampling models too. The core is machine-checked (Sections 3–4).
- **Works in practice.** Transformers trained from scalar rewards alone learn to predict, at the per-step cost of NTP (Section 5).
- **Learns without demonstrations.** With a proof checker as its only teacher, a 30M-parameter model goes from 20 to 86 solved miniF2F test problems, at 3.7 verifier calls each (Section 6).
- **A new road for language models.** One objective, a proper data reward plus task rewards, replaces the hand-off between pretraining and fine-tuning (Section 7).

1 Introduction

Every modern language model is trained twice. First it reads: billions of tokens of text, each one an exercise in guessing what comes next, scored by cross-entropy [1, 2]. Then it acts: it writes

answers, programs or proofs, and is rewarded for the good ones [3, 4, 5]. The two stages are taught, implemented and discussed as two different things, supervised learning and reinforcement learning, with different losses, different codebases and different intuitions. The second is often said to be the more general: reward maximization has been proposed as a principle sufficient for intelligence itself [6].

This paper takes that suggestion literally and pushes it as far as mathematics, machine-checked proofs and experiments allow, all of which run on a single laptop GPU. The title borrows the cadence of *Attention Is All You Need* [7], and it is meant in the precise sense stated in the box below.

Thesis

T1. NTP is a special case of RL. With the next-token distribution as action and a strictly proper scoring rule as reward, regret equals prediction error exactly; with the logarithmic score the RL gradient is the NTP gradient. NTP is RL with the variance taken out.

T2. RL therefore inherits universal prediction. One policy has finite cumulative regret on every prediction task generated by a computable source (Solomonoff, Hutter).

T3. The reduction survives in the token format language models use, provided rewards are computed from at least two samples; with one sample it provably fails.

T4. The same framework contains the rewards that NTP cannot use, such as a proof checker, so that an entire language model can in principle be specified by a single reward-maximization objective.

T1–T3 are theorems. T4 is a design principle, which we demonstrate with a working prototype.

Why this matters. If T1 holds, the line between "pretraining" and "RL fine-tuning" is a line between two reward signals, not between two paradigms: a dense, low-variance reward computed from data, and sparse rewards computed by verifiers, people or environments. The question "should we train language models with RL instead of NTP?" then dissolves into a more productive one: *which rewards, estimated how, and mixed in what proportion?* Several known pathologies of RL fine-tuning, most notably the collapse of output diversity, become predictable consequences of choosing a reward that is not proper (Section 4), and a principled remedy follows: keep a proper data reward in the objective. We call a model trained under one such objective an *RL-native language model*, and Section 7 lays out the road toward one.

Standing on classical shoulders. The universality statement (T1–T2) rests on classical work by Solomonoff [8, 9] and Hutter [10, 11, 12], which deserves to be far better known in the language-model community. What this paper adds is the exact link to how language models are trained today: the token-level analysis (T3), experiments with real Transformers and a real proof checker, and a concrete design for RL-native models. Each formal section opens with its idea in plain words, so the paper can be read by non-specialists.

Roadmap. Section 2 recalls the two paradigms and the tools that connect them. Sections 3 and 4 prove T1–T3. Sections 5 and 6 put the theory to work: training Transformers from scalar rewards alone, and learning to prove theorems from a verifier. Section 7 describes the road to RL-native language models, Section 8 states the precise scope of the claim, and Section 9 concludes.

2 Two Paradigms and the Bridge Between Them

Notation. $\mathcal{X}$ is a finite token alphabet with $m = |\mathcal{X}| \geq 2$ and $\Delta(\mathcal{X})$ the probability simplex in $\mathbb{R}^{\mathcal{X}}$; e_x is the one-hot vector of x. A *source* μ assigns to every history $x_{<t} = x_1 \cdots x_{t-1}$ a next-token distribution $\mu_t = \mu(\cdot \mid x_{<t})$; it induces a measure on infinite sequences and $\mathbb{E}_\mu$ denotes expectation under it. $\mathcal{M}_{\text{comp}}$ is the set of computable sources.

Next-token prediction. A predictor q maps histories to distributions $q_t \in \Delta(\mathcal{X})$ and is trained to minimize $\mathbb{E}_\mu \sum_t -\ln q_t(X_t)$, the cross-entropy, whose excess over the best possible value is the cumulative divergence $\sum_t \mathbb{E}_\mu \mathrm{KL}(\mu_t \| q_t)$ [13].

Reinforcement learning. An agent observes a history, takes an action, and receives a reward; a policy maps histories to (distributions over) actions and is judged by its expected total reward, or equivalently by its regret against a benchmark policy [14]. Policy-gradient methods estimate the gradient of expected reward from sampled actions and scalar rewards alone [15].

The bridge: proper scoring rules. A scoring rule $S(q, x)$ rewards a forecast $q \in \Delta(\mathcal{X})$ after outcome x is revealed. Its expected value under p is $S(q; p) = \sum_x p(x) S(q, x)$. It is *proper* if $S(p; p) \geq S(q; p)$ for all p, q, and *strictly proper* if equality forces $q = p$ [16, 17]; in words, honesty is the unique best strategy. The gap $d_S(p, q) = S(p; p) - S(q; p) \geq 0$ is the rule's divergence. Two rules matter here:

$$\text{logarithmic:} \quad S_L(q, x) = \ln q(x), \qquad d_{S_L}(p, q) = \mathrm{KL}(p \,\|\, q), \tag{1}$$

$$\text{Brier [18]:} \quad S_B(q, x) = 1 - \tfrac{1}{2}\|q - e_x\|_2^2 \in [0, 1], \qquad d_{S_B}(p, q) = \tfrac{1}{2}\|q - p\|_2^2. \tag{2}$$

The Brier identity follows from $\mathbb{E}_{X \sim p}\|q - e_X\|^2 = \|q - p\|^2 + \mathbb{E}_{X \sim p}\|p - e_X\|^2$, the cross term vanishing because $\mathbb{E}_{X \sim p} e_X = p$.

Universal prediction. Let $\mathcal{M}$ be a countable class of sources with prior weights $w_\nu > 0$, $\sum_\nu w_\nu \leq 1$. The Bayes mixture $\xi = \sum_\nu w_\nu \nu / \sum_\nu w_\nu$ dominates every member: $\xi(x_{1:n}) \geq w_\mu \mu(x_{1:n})$.

Theorem 1 (Solomonoff [9], Hutter [10])**.** *If $\xi(x_{1:n}) \geq c\,\mu(x_{1:n})$ for all $n, x_{1:n}$ and some $c > 0$, then for every n*

$$\sum_{t=1}^{n} \mathbb{E}_\mu \|\xi_t - \mu_t\|_2^2 \;\leq\; \sum_{t=1}^{n} \mathbb{E}_\mu \mathrm{KL}(\mu_t \| \xi_t) \;\leq\; \ln \frac{1}{c},$$

and $\xi_t - \mu_t \to 0$ μ-almost surely.

Taking for $\mathcal{M}$ all lower semicomputable semimeasures with weights $2^{-K(\nu)}$ (K the prefix Kolmogorov complexity) gives Solomonoff's universal predictor M; its normalization M^{norm} is a proper source that still dominates every computable μ with $c = 2^{-K(\mu) - O(1)}$ [19, 20], so

$$\sum_{t=1}^{\infty} \mathbb{E}_\mu \|M_t^{\text{norm}} - \mu_t\|_2^2 \;\leq\; (\ln 2)\, K(\mu) + O(1) \qquad \text{for every } \mu \in \mathcal{M}_{\text{comp}}. \tag{3}$$

One predictor works for every computable source; the price of a source is its description length. The predictor is incomputable. A proof of Theorem 1 is recalled in Appendix A.

3 Next-Token Prediction Is Reinforcement Learning

The idea in words

Turn "guess the next word" into a game. On each turn the player does not name a word; it hands in a betting slip listing a probability for every word. The game then reveals the real next word and pays according to a proper scoring rule. Because the rule is proper, the only way to maximize expected pay is to bet one's true beliefs, and the pay lost relative to a perfectly informed player is exactly the distance between the slip and the truth. Because the player's bets never change what text comes next, there is nothing to explore and nothing to plan: playing well today never costs anything tomorrow. Playing this game well and predicting well are the same skill.

Definition 1 (Prediction environment)**.** Fix a source μ and a scoring rule S. In the environment E_μ^S, at step t the agent observes $x_{<t}$, outputs an action $q_t \in \Delta(\mathcal{X})$, the environment draws $X_t \sim \mu_t$ independently of q_t, reveals it, and pays $S(q_t, X_t)$. The regret of a policy π is $R_T(\pi, \mu) = \mathbb{E}_\mu \sum_{t \le T}[S(\mu_t, X_t) - S(q_t, X_t)]$, measured against the informed policy that outputs μ_t.

Lemma 1 (Regret equals prediction error)**.** *For every policy, source and horizon, $R_T(\pi, \mu) = \sum_{t \le T} \mathbb{E}_\mu \, d_S(\mu_t, q_t)$. For the Brier reward $R_T = \frac{1}{2} \sum_t \mathbb{E}_\mu \|q_t - \mu_t\|^2$; for the log reward $R_T = \sum_t \mathbb{E}_\mu \mathrm{KL}(\mu_t \| q_t)$. If S is proper the informed policy is optimal among all policies, at every horizon.*

Proof. Condition on $x_{<t}$: then q_t is fixed and $X_t \sim \mu_t$, so the conditional expected regret of step t is $d_S(\mu_t, q_t)$. Sum and take expectations. Actions do not influence observations, so step-wise optimality is global optimality. □

Corollary 1 (NTP is RL with the variance taken out)**.** *Let $q_t = \mathrm{softmax}(z_\theta(x_{<t}))$. In $E_\mu^{S_L}$ the expected reward of step t given the observed token is $\ln q_t(x_t)$, which is minus the NTP loss. Its gradient is the cross-entropy gradient; no sampling of actions is involved because the action is the whole distribution and the reward is differentiable in it.*

The corollary is elementary, and that is the point. In the vocabulary of RL, ordinary pretraining is an *exact* policy-gradient method in a non-reactive environment: the reward of every possible action is known in closed form once the next token is revealed, so the expectation over actions can be taken analytically. Sampling-based RL estimates the same quantity with noise. Section 5 measures how large that noise is in practice.

Theorem 2 (Universality in prediction environments)**.** *Let π^ξ output the Bayes mixture prediction ξ_t. For every $\mu \in \mathcal{M}$, with the Brier reward $\sup_T R_T(\pi^\xi, \mu) \le \frac{1}{2} \ln(1/w_\mu)$, and with the log reward $R_T(\pi^\xi, \mu) \le \ln(1/w_\mu)$ for all T. The policy that outputs M_t^{norm} satisfies, under the Brier reward,*

$$\exists \pi \;\; \forall \mu \in \mathcal{M}_{\mathrm{comp}} \;\; \forall T : \qquad R_T(\pi, \mu) \le \tfrac{1}{2}(\ln 2) K(\mu) + O(1).$$

Proof. Lemma 1 and Theorem 1 with (3). □

Remark 1 (Reading the quantifiers)**.** One policy, chosen before any data is seen, works for every computable source; the bound depends on the source but not on the horizon. Bounded cumulative regret means that the average regret vanishes and predictions converge, not that the policy is right from the start. The policy is incomputable: the theorem is an existence result about what the RL framework can express and achieve, not about what a particular algorithm will find. Hutter already analyzed the universal mixture as a decision maker under general loss in non-reactive environments [11] and treated sequence prediction as a special case of AIXI [12]; Theorem 2 adds only the exact regret identity that makes the transfer loss-free.

4 When Actions Are Tokens

The idea in words

A language model does not hand in a betting slip; it *samples* a token. If the reward looks at one sampled token only, then the expected reward is an average of the rewards of individual tokens, and the best strategy is always to put all probability on the single best token. No reward design, however clever, can make "tell the truth about the distribution" optimal. Two samples change this: comparing them reveals how concentrated the policy is, and a reward that subtracts their agreement turns honesty back into the optimum.

Definition 2 (K-sample token environment)**.** At step t the policy specifies $\pi_t \in \Delta(\mathcal{X})$, the agent draws $A_1, \dots, A_K \overset{\text{iid}}{\sim} \pi_t$, the environment draws $X_t \sim \mu_t$, and pays $\rho(A_{1:K}, X_t)$. Write $U_\rho(q;p) = \mathbb{E}_{A\sim q^{\otimes K}, X\sim p}\,\rho(A,X)$. Lemma 1 holds verbatim with $d(p,q) = U_\rho(p;p) - U_\rho(q;p)$.

Proposition 1 (One sample is never enough)**.** *For $K = 1$, $U_\rho(q;p) = \sum_a q(a)\sum_x p(x)\rho(a,x)$ is linear in q. For every ρ and every p with two or more support points, some point mass does at least as well as p; p is never the unique optimum.*

What if a regularizer of the policy is added, as practitioners do with entropy bonuses and KL penalties? The answer is sharp.

Theorem 3 (Only quadratic objectives are reachable)**.** *Let $R_{ax} = \rho(a,x)$ and let $\Omega : \Delta(\mathcal{X}) \to \mathbb{R}$ be continuous, and differentiable on the relative interior along the tangent space $T_0 = \{v : \sum_a v(a) = 0\}$. If every interior p maximizes $J_p(q) = U_\rho(q;p) + \Omega(q)$, then $J_p(p) - J_p(q) = \frac{1}{2}(q-p)^\top R(q-p)$ for all interior p and all q. Hence the induced divergence is a fixed quadratic form, Ω is quadratic up to an affine term, and no entropy or KL regularizer, and no single-sample reward with any smooth regularizer, yields the logarithmic score.*

Proposition 2 (Two samples suffice)**.** *For $K = 2$ and $\rho_2(a,a',x) = \mathbf{1}\{a = x\} + \mathbf{1}\{a' = x\} - \mathbf{1}\{a = a'\}$,*

$$U_{\rho_2}(q;p) = 2\langle q,p\rangle - \|q\|^2 = \|p\|^2 - \|q-p\|^2.$$

The reward is strictly proper on the whole simplex and its regret is exactly $\|q-p\|^2$.

Proof. $\mathbb{E}\mathbf{1}\{A = X\} = \mathbb{E}\mathbf{1}\{A' = X\} = \langle q,p\rangle$ and $\mathbb{E}\mathbf{1}\{A = A'\} = \|q\|^2$ by independence. □

The reward ρ_2 is the fair Brier score of Ferro [21] for a two-member ensemble. Its policy gradient uses only sampled tokens: $\nabla U_{\rho_2} = \mathbb{E}[\rho_2(A,A',X)(\nabla \ln \pi(A) + \nabla \ln \pi(A'))]$.

Theorem 4 (Universality for token-sampling policies)**.** *In the two-sample token environment with reward ρ_2, the policy that samples from ξ_t has $\sup_T R_T \le \ln(1/w_\mu)$ for every $\mu \in \mathcal{M}$, and the policy that samples from M_t^{norm} has $\sup_T R_T \le (\ln 2)K(\mu) + O(1)$ for every computable μ.*

Groups: fair versus biased. Group-based methods such as GRPO [4] draw K samples per context. Estimating $\|q\|^2$ by collisions among all K^2 ordered pairs, diagonal included, is the natural "plug in the empirical distribution" choice. It is biased:

Proposition 3 (Biased group rewards sharpen and truncate)**.** *With $c = (K-1)/K$, the biased (V-statistic) group reward has $U(q;p) = 2\langle q,p\rangle - c\|q\|^2 - 1/K$, whose unique maximizer is the Euclidean projection of p/c onto the simplex: $q^\star(a) = (\frac{K}{K-1}p(a) - \tau)_+$. It increases every likelihood ratio between surviving tokens and assigns zero probability to every token with $p(a) \le c\tau$.*

For $K = 2$ and $p = (0.5, 0.3, 0.15, 0.05)$ the biased optimum is $(0.7, 0.3, 0, 0)$: half of the vocabulary disappears. The fair (U-statistic) version, which leaves out the diagonal, has $U = \|p\|^2 - \|q-p\|^2$ for every $K \ge 2$.

Proposition 4 (No finite number of samples reaches the log score)**.** *For any K and ρ, $U_\rho(q;p)$ is a polynomial of degree at most K in q, so its divergence is bounded on $\Delta(\mathcal{X}) \times \Delta(\mathcal{X})$ and is never a positive multiple of* $\mathrm{KL}(p\|q)$*.*

Proposition 4 has a practical reading that Section 5 confirms: a learner that only ever sees sampled tokens and scalar rewards is confined to Brier-type objectives, and pays whatever Brier-type objectives cost relative to the logarithmic score. Proofs of Theorem 3 and Proposition 3 are in Appendix A.

A lesson for RL fine-tuning. RL with verifiable rewards scores one sampled response, so its objective is linear in the policy and its optima are concentrated on reward-maximizing responses (Proposition 1); with a KL penalty the optimum is a reweighting $\pi^\star \propto \pi_{\mathrm{ref}} e^{r/\beta}$ of the reference [22, 23, 24]. Loss of diversity is thus the expected behavior of these objectives, not an accident of training, consistent with the empirical finding that RL-trained models can be overtaken by their base models at large pass@k [25]. The theory also names the remedy: keep a strictly proper reward on the data in the objective. Section 6 tests this.

Machine-checked. The algebraic core of Sections 3 and 4, including the regret identities and the one- and two-sample results, is formally verified in Lean 4 with Mathlib [26, 27]; Solomonoff's bound enters as a stated assumption.

5 Experiment I: Training a Transformer from Rewards Alone

The question

The theorems say that an RL agent *can* predict as well as anything. Does an ordinary Transformer, trained by ordinary policy gradient from nothing but scalar rewards, actually learn to predict? And what does it cost compared with NTP?

Training signals. The model is a standard decoder-only Transformer; only the training signal changes. In every RL mode the true next token is hidden inside the environment and used only to compute a scalar reward; no gradient flows from it. We compare `ntp` (cross-entropy); `brier` (action = the whole distribution, reward S_B, differentiable in the action); `pg`K (action = K sampled tokens, fair group reward, REINFORCE with a batch baseline and per-sample credit whose expectation is exactly $\nabla(2\langle q,p\rangle - \|q\|^2)$); `pgv`$K$ (the biased group reward); and `pg1` (one sampled token, reward $\mathbf{1}\{A = x\}$, optionally with an entropy bonus).

Three settings. (i) A synthetic order-2 Markov source over 16 tokens with Dirichlet(0.5) transition rows, where the true μ_t is known and the distance to it can be measured; 4 layers, width 128, 4000 steps. (ii) Character-level Lean 4 code: a 9 MB subset of Mathlib, 313 symbols, 6 layers, width 256, about 5M parameters, 4000 steps. (iii) BPE-level Lean: all of Mathlib, a byte-level BPE vocabulary of 8192, 30.6M training tokens with held-out files for validation, 8 layers, width 512, 29.7M parameters, 15,000 steps of 32×512 tokens. All runs use one laptop GPU (RTX PRO 4000 Blackwell, 16 GB) and one seed.

Result 1: rewards alone suffice to learn the distribution. Table 1 confirms every prediction of Section 4 on the synthetic source. The fair sampled-token reward recovers the true conditional distribution to within $\|q - \mu\|^2 = 0.0014$ (NTP: 0.0006); the single-sample reward collapses; the entropy bonus gets the mode right and the distribution wrong; the biased group reward sharpens. On real code the RL-trained models produce recognizable Lean (declarations, `by simp`, `rfl`, attributes), and their validation loss tracks NTP's at a distance (Fig. 1).

Table 1: Synthetic source with known truth (order-2 Markov, 16 tokens, 4000 steps). Mode accuracy is the fraction of contexts in which the model's most likely token is the true most likely token.

Method	Learner sees	$\|q-\mu\|^2$	KL($\mu\|q$)	Mode acc.	Theory
ntp	true token	0.0006	0.005	0.95	baseline
brier	scalar reward	0.0010	0.027	0.95	proper (Lemma 1)
pg8, fair	scalar reward	0.0014	0.033	0.95	proper (Prop. 2)
pg2, fair	scalar reward	0.0041	0.074	0.92	proper, noisier
pgv2, biased	scalar reward	0.0574	0.532	0.93	sharpens (Prop. 3)
pg1	scalar reward	0.903	9.74	0.19	collapses (Prop. 1)
pg1 + entropy 0.05	scalar reward	0.219	0.813	0.96	mode only (Thm. 3)

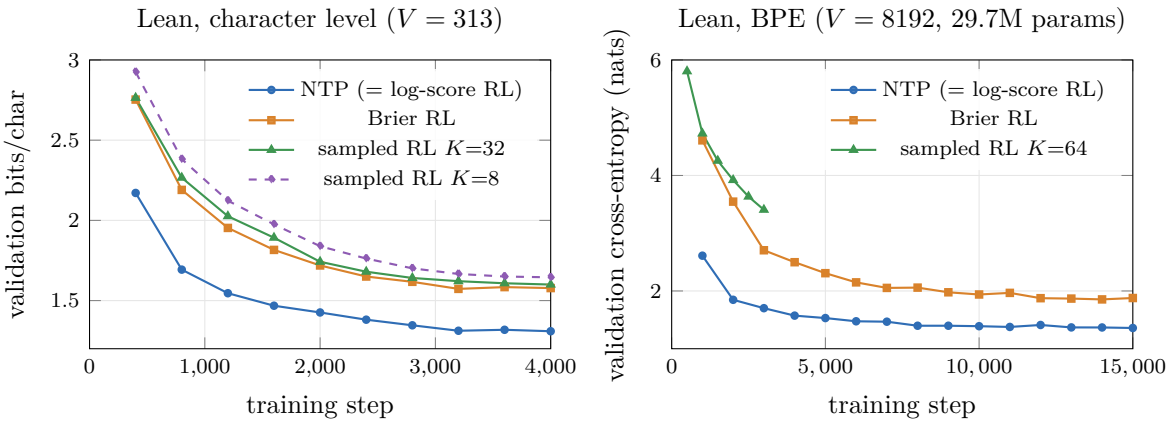


Figure 1: Validation loss of Transformers trained from rewards alone, compared with NTP. All curves descend; the RL curves lag by a roughly constant factor in steps that grows with the vocabulary.

Result 2: the price is statistical, not computational. Per training step, RL and NTP cost the same: the forward and backward passes dominate, and sampling K tokens, counting collisions and computing $\|q\|^2$ are cheap (Table 2). The difference lies in how much each step teaches. Table 3 reports, for each RL run, the number of steps NTP needed to reach the same quality (by interpolating the NTP curve), and the ratio.

Result 3: the gap is mostly the scoring rule, as Proposition 4 predicts. The ratios decompose. Comparing brier with NTP isolates the choice of scoring rule (Brier versus logarithmic) with no sampling at all: 3.6–3.7× at small vocabularies and 7.1× at $V = 8192$. Comparing pg$K$ with brier isolates sampling noise: an extra 1.05–1.25× for $V \leq 313$, decreasing with K, and 1.4× at $V = 8192$ with $K = 64$ (Brier reaches the cross-entropy that pg64 attains at step 3000 after about 2130 steps). The scoring-rule term grows with the vocabulary for a simple reason: the logit gradient of the log score on the correct token is $1 - q(x)$, whereas that of the Brier score is $q(x)(1 - 2q(x) + \|q\|^2) \approx q(x)$, which vanishes when $q(x)$ is small, and early in training $q(x) \approx 1/V$. At BPE scale this is decisive: at every step the NTP model scores higher than the Brier-trained model *even on the Brier score* (0.284 versus 0.273 at step 15,000). The metric also matters: on squared error, the Brier rule's own divergence, Brier-RL needs only 1.3× the NTP steps on the synthetic source. Since sampled-token learners are confined to polynomial, Brier-type objectives (Proposition 4), they inherit this price.

Table 2: Per-step cost relative to NTP (median of three interleaved rounds on a shared GPU; ratios are the meaningful quantity).

Scale	Method	Time per step	Peak memory
Character level, $V = 313$	brier	0.93×	1.04×
	pg $K = 8$ / 32	0.95× / 0.94×	1.04× / 1.05×
	pg $K = 128$	1.11×	1.66×
BPE, $V = 8192$	brier	1.15×	1.12×
	pg $K = 8$ / 64	1.13× / 1.21×	1.44× / 1.44×

Table 3: Statistical efficiency: steps an RL run took, steps NTP needed to reach the same value, and their ratio.

Setting	Metric	Method	RL steps	NTP-equivalent	Ratio
Synthetic, $V = 16$	$\mathrm{KL}(\mu\|q)$	brier	4000	1077	3.7
		pg8	4000	870	4.6
		pg2	4000	549	7.3
	$\|q - \mu\|^2$	brier / pg8	4000	3196 / 2932	1.3 / 1.4
Lean char, $V = 313$	bits/char	brier	4000	1126	3.6
		pg32	4000	1052	3.8
		pg8	4000	931	4.3
Lean BPE, $V = 8192$	cross-entropy	brier	15000	2117	7.1
		pg64	3000	578	5.2

What Experiment I shows

RL from scalar rewards does learn next-token prediction in a real Transformer, at the same cost per step. The efficient instance of this RL is the one that takes the expectation over actions analytically with the logarithmic score, which is NTP itself. Any RL-native design should therefore keep this instance as its data reward and spend sampling only where no analytic expectation exists.

6 Experiment II: Rewards That NTP Cannot Use

The question

If NTP is the dense, analytic corner of RL, what does the rest of RL buy? The clearest case is a reward that has no corpus behind it: a proof checker. NTP can only imitate proofs that already exist; RL can learn from the checker alone. Can a small model that has never seen a proof of the target problems learn to produce them, and what goes wrong?

Setup. Each experiment starts from a model pretrained on Lean code (Section 5). For each statement the model samples tactic proofs; Lean checks them; the reward is 1 if Lean accepts the proof and 0 otherwise, with group-normalized advantages. Optionally the stage-1 proper reward on corpus text is kept as an *anchor* with weight λ, so that the objective is a single mixture of a data reward and a verifier reward. No proof examples are ever shown.

Toy propositions. With the character-level model, randomly generated core-Lean goals from 12 templates (introduction and elimination of $\wedge, \vee, \to$, simple arithmetic) and one-line proofs, pass@1 on 48 held-out goals rose from 1.3% to 22.9% with the verifier reward alone, and to

52.1% with the fair-Brier anchor ($\lambda = 0.5$). Without the anchor the policy collapsed onto a single answer (`simp`) within 50 steps and stayed there; with it, the model learned to cite the relevant hypothesis (`simp [h2]`), producing nine distinct correct proof forms. This is Section 4 in action: a pure 0–1 task reward is mode-seeking, and a proper data reward preserves the diversity that further learning needs.

miniF2F, one line. With the BPE model we used the Lean 4 statements of miniF2F [28] distributed with DeepSeek-Prover-V1.5 [29]; 423 of 488 elaborate under our Mathlib version. RL runs on the `valid` split and evaluation on the 212 `test` problems. The reference point is a strong automation baseline: trying each of ten tactics (`grind`, `nlinarith`, `omega`, `linarith`, `ring_nf`, `aesop`, `norm_num`, `decide`, `simp`, `positivity`) solves 89 test problems. Before RL the model solves 23 (16 samples per problem). After 60 RL steps it solves 76–80 with the verifier reward and the anchor; adding larger groups and a diversity reward lifts this to 84 on both seeds (Table 4). Across all runs RL solves 88 problems, three of them beyond the baseline through tactic combinations such as `ring <;> norm_num <;> grind`.

Reward hacking appears exactly where the reward is not proper. Our diversity bonus first rewarded textually distinct correct proofs; the policy learned to pad `grind` with irrelevant lemma hints (`grind [add_assoc]`, `grind [mul_comm]`). Rewarding distinct tactic sequences instead, it learned to repeat tactics (`grind <;> grind <;> grind`). Both are instances of reward hacking [30]: a diversity measure defined on the surface form can be satisfied by changing the surface form. A strictly proper data reward is immune to this particular failure, because its expectation is maximized only by the true distribution, whatever the surface; this is a practical argument for proper scores as the diversity-preserving term.

miniF2F, multi-line, with goal states. Finally we let the model write proofs step by step. After each step Lean reports the remaining goals, which are shown to the model as a comment; best-first search orders partial proofs by cumulative log-probability and discards steps that do not change the goal. RL is expert iteration [31, 32]: search on the `valid` split, reward every step of every accepted proof, raise its probability under the anchor, repeat twice. Table 4 and Fig. 2 report the results. The search with automation tactics as additional candidates solves 94 problems, above the 89 baseline. A search over the same ten tactics with no model also solves 94, and the five problems beyond the baseline are two-tactic combinations such as `aesop` followed by `nlinarith`, so at this scale the extra problems come from search. What RL adds is efficiency, and a large amount of it: searching with the RL-trained policy alone solves 86 problems with 3.7 Lean calls per problem on average, against 20 before RL; the ten-tactic baseline needs 10 calls for 89 and the tactic-only search 20.3 calls for 94. With automation candidates added, RL halves the verifier calls (47.0 → 21.6) and the wall-clock time (24 → 9 minutes) at the same solve rate.

What Experiment II shows

A verifier is a reward NTP cannot use, and RL from it turns a model that has never seen a proof into a competent, cheap prover, with no demonstrations. Pure task rewards collapse the policy; a proper data reward in the same objective counteracts collapse and, unlike surface-level diversity bonuses, cannot be gamed by rewriting. At only 30M parameters the learned policy already recovers, and makes about three times cheaper, the competence of a strong automation baseline; going beyond it is a matter of scale.

Table 4: miniF2F test (212 problems), Lean 4 and Mathlib. "Beyond" counts problems the ten-tactic baseline does not solve; "calls" is the mean number of Lean checks per problem. One-line rows: mean over two seeds, 60 RL steps, 16 samples per problem at evaluation.

Method	Solved	Beyond	Calls
Ten automation tactics, each tried once (baseline)	89	–	10
Tactic-only best-first search, no model	94	5	20.3
One-line proofs, sampled			
pretrained model, no RL	23	0	16
RL: verifier reward + anchor $\lambda = 0.5$	78	0	16
RL: + larger groups, stronger anchor, diversity reward	84	0.5	16
Multi-line proofs, best-first search with goal states			
pretrained model, policy only	20	0	14.9
RL (expert iteration), policy only	86	2	**3.7**
pretrained model + automation candidates	93	5	47.0
RL + automation candidates	94	5	21.6

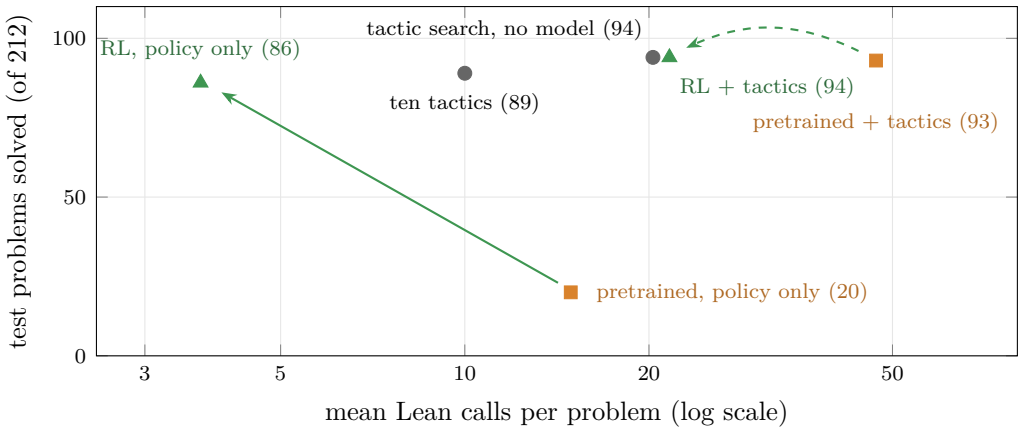


Figure 2: Solve rate against verifier cost on miniF2F test. Arrows show the effect of RL from the verifier. RL turns a model that cannot prove into one that reaches 97% of the automation baseline at a third of its cost; at this model scale the solve rate is bounded by what search over automation achieves, and larger models are the natural next step.

7 The Road Ahead: RL-Native Language Models

The usual pipeline treats a language model as a predictor first and an agent second. The results above suggest treating it as an agent throughout, with prediction as the agent's densest reward. We call this the RL-native view and state it as one objective.

One objective. For a policy π_θ over token sequences, a data distribution $\mathcal{D}$, a strictly proper scoring rule S and task rewards r_j with environments $\mathcal{E}_j$,

$$J(\theta) \;=\; \lambda \underbrace{\mathbb{E}_{x\sim\mathcal{D}} \sum_t S\big(\pi_\theta(\cdot \mid x_{<t}), x_t\big)}_{\text{data reward: analytic, dense, proper}} \;+\; \sum_j \beta_j \underbrace{\mathbb{E}_{e\sim\mathcal{E}_j,\, y\sim\pi_\theta(\cdot|e)}\, r_j(e,y)}_{\text{task rewards: sampled, sparse}} . \tag{4}$$

With $S = S_L$ and $\beta_j = 0$ this is pretraining. With $\lambda = 0$ it is RL with verifiable rewards and its collapse. Everything in between is a single reward-maximization problem, with one schedule for λ and β_j in place of a hand-off between stages. Section 4 explains why the data term should stay ($\lambda > 0$): it is the only term in (4) whose optimum is a distribution rather than a mode,

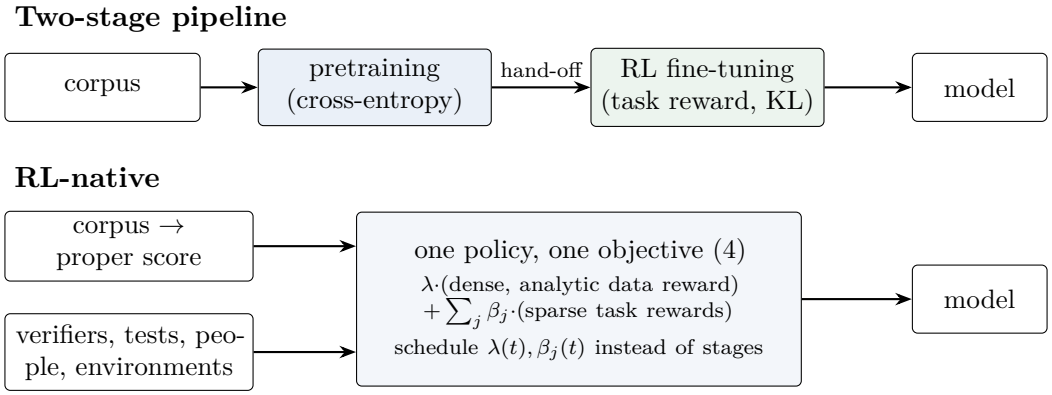


Figure 3: The two-stage pipeline and the RL-native view. In the latter, pretraining is not a separate paradigm but the densest, analytically computable reward in a single objective.

and Experiment II shows that it is the term that cannot be gamed by surface rewriting. Fig. 3 contrasts the two views.

Milestones on the road. The analysis and experiments point to seven concrete problems whose solution would turn the RL-native view into practice. We list them in rough order of urgency.

P1. Keep the analytic corner. Wherever an expectation over actions can be computed in closed form, as it can for next-token rewards on data, compute it, and use the log score. Experiment I shows that giving this up costs 3.6–7.1× in steps. Sampling should be reserved for rewards that only exist as black boxes.

P2. Variance and vocabulary. Sampled-token learning degrades as the vocabulary grows, because the chance that a sample hits the observed token is small. Control variates that exploit the known structure of (4), importance sampling from a proposal concentrated near the data token, and hierarchical or factorized action spaces are natural directions, as is a theory of how many samples per context a fair group reward needs as a function of the entropy of μ_t.

P3. Proper rewards at the level of responses. Our token-level results lift directly only to rewards on whole responses when those rewards are themselves proper scores of a distribution. Designing sequence-level fair rewards that are cheap to estimate from a group of sampled responses, and whose optimum is a calibrated distribution over correct answers rather than one answer, is the response-level analogue of Proposition 2.

P4. Semantic, not syntactic, diversity. Experiment II shows that diversity defined on surface forms is gamed immediately. Diversity measured on meaning (for proofs: proof terms or goal trajectories; for programs: behavior on tests) is needed wherever a proper data reward is unavailable.

P5. Verifiable environments as the new corpus. If pretraining is the data reward, then curated environments with checkable rewards, such as formal mathematics, program synthesis and simulators, play the role that text corpora play today. Their scale, diversity and curricula become first-class research objects [32, 33].

P6. Exploration and irreversibility. All guarantees above live in non-reactive environments. In interactive settings with irreversible actions no policy matches the informed optimum (Section 8); RL-native models that act in the world need weaker, history-relative optimality notions and safe exploration.

Table 5: Readings of "reinforcement learning is all you need" and their status.

Reading	Status
RL can express every probabilistic prediction task, NTP included	True with a strictly proper reward (Lemma 1, Corollary 1)
One policy has bounded regret on all computable prediction environments	True, incomputable policy (Theorems 2, 4)
Same, with single-sample token rewards	False (Proposition 1, Theorem 3)
Ordinary policy gradient learns NTP from rewards alone	Observed at small scale, 3.6–7.1× slower in steps (Section 5)
RL from a verifier learns skills without demonstrations	Observed at small scale; reaches the automation baseline at a third of the cost (Section 6)
One policy is near-optimal in all computable interactive environments	False against informed benchmarks (Proposition 5); weaker notions partly true [34, 36]
RL-native training beats the two-stage pipeline at scale	Open; the next frontier (Section 7)

P7. Scaling laws for the mixture. How should λ and β_j evolve with compute? Our data suggest that early training should be dominated by the analytic data reward, which is most informative per step, and that sparse rewards pay off once the prior places non-negligible probability on rewarded behavior, as the jump from 20 to 86 problems illustrates.

8 Precise Scope of the Claim

"All you need" is a claim about sufficiency in principle, and its strength comes from being exact. This section states where it holds and where the frontier lies.

Proposition 5 (No universal policy against informed optimal policies)**.** *Let E_L, E_R be two computable environments that look identical before the first action; in E_L choosing left leads to an absorbing state with reward 1 forever and right to one with reward 0; E_R is the mirror image. A policy choosing left with probability θ earns average reward θ in E_L and $1-\theta$ in E_R, while the informed optimal policy earns 1 in both; in one of them the regret grows at least as $T/2$.*

The obstruction is informational and holds for incomputable policies too [34, 35]. Prediction environments escape it because a wrong prediction costs reward but never information. Table 5 summarizes what is and is not established.

Scale and practice. The universal policy is incomputable; the universality theorems establish what the RL framework can achieve, not which policy a given algorithm finds. Our experiments use models of 5–30M parameters on one GPU, mostly with one seed, and the miniF2F gains are in cost rather than solve rate; testing the approach at larger scale is the natural next step. Finally, Corollary 1 is a statement of continuity, not of replacement: for data rewards the best RL algorithm *is* NTP, so the RL-native view keeps everything NTP does well and adds to it the rewards that NTP cannot use.

9 Conclusion

Next-token prediction and reinforcement learning are not two paradigms. With a strictly proper scoring rule as reward, the first is the special case of the second in which the environment ignores the agent and the expectation over actions can be taken exactly; the familiar cross-entropy loss

is the policy gradient of the logarithmic score. Through this identity the RL framework inherits Solomonoff's universality on prediction tasks, and with a fair two-sample reward it does so even in the token-sampling format that language models use. Ordinary Transformers trained from scalar rewards alone do learn to predict, at the cost per step of NTP and with a statistical price that the theory traces to the scoring rule. And the same framework holds the rewards that imitation cannot use: a proof checker alone taught a small model, shown no proofs, to prove competition problems cheaply, with a proper data reward guarding it against collapse.

In this precise sense, reinforcement learning is all you need: one reward-maximization objective can specify the whole training of a language model, with pretraining as its densest reward and verifiers, tests and environments as the rest. This opens a second road for language models, alongside the familiar one of imitating ever more text: models that learn, from the start, from every signal that can tell them how well they are doing. Section 7 maps the milestones along that road.

A Proofs

Theorem 1. Dominance gives $\ln(\mu^{(n)}/\xi^{(n)}) \le \ln(1/c)$ pointwise on the support of $\mu^{(n)}$, hence $\mathrm{KL}(\mu^{(n)}\|\xi^{(n)}) \le \ln(1/c)$. The chain rule gives $\mathrm{KL}(\mu^{(n)}\|\xi^{(n)}) = \sum_{t\le n} \mathbb{E}_\mu \mathrm{KL}(\mu_t\|\xi_t)$. For $p, q \in \Delta(\mathcal{X})$, $\|p-q\|_2^2 \le \frac{1}{2}\|p-q\|_1^2 \le \mathrm{KL}(p\|q)$ by Pinsker's inequality [37]. The partial sums of $\|\xi_t - \mu_t\|^2$ are non-decreasing with bounded expectations, so the infinite sum is finite almost surely and its terms tend to zero.

Theorem 3. Let $g(z)$ be the gradient of Ω at interior z along T_0. Since interior p maximizes the differentiable J_p, its directional derivatives vanish: $\langle Rp + g(p), v\rangle = 0$ for $v \in T_0$, i.e. $\langle g(z), v\rangle = -v^\top Rz$ for all interior z. For interior p, q and $d = q - p$, $\varphi(s) = \Omega(p + sd)$ has $\varphi'(s) = -d^\top R(p + sd)$, so $\Omega(q) - \Omega(p) = -d^\top Rp - \frac{1}{2}d^\top Rd$. Since $U_\rho(q;p) - U_\rho(p;p) = d^\top Rp$, adding gives $J_p(q) - J_p(p) = -\frac{1}{2}d^\top Rd$; continuity extends this to $q \in \Delta(\mathcal{X})$. For $\Omega = \beta H$ the second derivative along $e_a - e_b$ is $-\beta(1/q(a) + 1/q(b))$, which is not constant, so entropy (and KL, which differs by a linear term) is excluded; and a quadratic divergence is never $\mathrm{KL}(p\|q)$.

Proposition 3. $\mathbb{E}\sum_{k,l} \mathbf{1}\{A_k = A_l\} = K + K(K-1)\|q\|^2$ gives the expression for U. Completing the square, $U = -c\|q - p/c\|^2 + \text{const}$, so the maximizer is the projection, which has the thresholding form with $\tau > 0$ because $\sum_a p(a)/c = 1/c > 1$. Subtracting $\tau > 0$ from $p(a)/c > p(b)/c > \tau$ strictly increases their ratio, and coordinates with $p(a)/c \le \tau$ are clipped to zero.

Proposition 4. $U_\rho(q;p) = \sum_{a_{1:K},x} q(a_1)\cdots q(a_K)\, p(x)\, \rho(a_{1:K}, x)$ is a polynomial, bounded on the compact set $\Delta(\mathcal{X}) \times \Delta(\mathcal{X})$, whereas $\mathrm{KL}(p\|q) \to \infty$ as $q(a) \to 0$ with $p(a) > 0$.

B Experimental Details

Models and optimization. Decoder-only Transformers with learned positional embeddings and pre-layer normalization. Character-level runs: AdamW, learning rate 10^{-3} with 100 warm-up steps and cosine decay, batch 64×128 (Lean) or 64×64 (synthetic), dropout 0.1 on text. BPE runs: width 512, 8 layers, 8 heads, context 512, tied embeddings, AdamW ($\beta = (0.9, 0.95)$, weight decay 0.1), learning rate 6×10^{-4} with 500 warm-up steps and cosine decay to 10%, batch 32×512, bfloat16 autocast. Sampled-token rewards use per-sample credit $r_k = 2\mathbf{1}\{A_k = x\} - w\,\#\{l \ne k : A_l = A_k\}$ with $w = 2/(K-1)$ (fair) or $2/K$ (biased) and a batch-mean baseline.

Verifier RL. Toy propositions: core Lean without Mathlib, one file per candidate, 8 goals $\times$ 8 samples per step, 200 steps, learning rate 10^{-4}. miniF2F one-line: 8 problems $\times$ 16 or 32

samples per step, 60 steps, learning rate 2×10^{-5}, anchor on 8 corpus windows per step; diversity reward shares a unit reward among identical correct proofs in a group and adds a one-time novelty bonus of 0.5. Multi-line: best-first search with 8 policy samples per expansion, at most 6 expansions and depth 6; steps that leave the goal unchanged or revisit a state are pruned; expert iteration uses two rounds of search on `valid` with at most 4 expansions, three epochs over the accepted steps, anchor $\lambda = 0.5$. Lean checks batch 16 candidates per file under `import Mathlib` with `maxHeartbeats 400000`; candidates containing comments, `sorry`, `admit`, `native_decide`, search commands or declaration keywords are rejected before checking, and a nine-case regression test guards error attribution.

Two practical pitfalls. Batching many candidate proofs into one Lean file lets a malformed candidate (for example, a lone comment after `by`) push its parse error onto the next theorem, so that it is scored as correct; error attribution must account for this, and comments and search commands such as `exact?` must be rejected. Byte-level BPE merges the newline after `by` with the following indentation, so a prompt that ends in a bare newline is out of distribution and must instead end at `by`. All reported results use the corrected verifier and prompt.

Hardware and software. One NVIDIA RTX PRO 4000 Blackwell laptop GPU (16 GB), 20 CPU cores, PyTorch 2.13, Lean 4.34.1 with Mathlib v4.34.1 for verification and Lean 4.35.0 for core-Lean checks.

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